

A Complete, Governed Research Workflow

Reviewer demonstration dossier from one real ICU analysis run

REVIEWER DEMONSTRATION COMPLETE

```
Run run_20260815T061842_5049c6 · authority=engineering_validation_only ·  
publication_authorized=false  
report_payload_sha256=7bac0ac70899b6b81e8f06840b8af41134ac8eb8067d1de43e307247f87b1ea0
```

Executive finding

This reviewer demonstration follows one exact plan from approval through execution, aggregate results, figures, provenance, and final authority adjudication. The clinical phenotype analysis is a bounded test case; the demonstrated contribution is the governed workflow.

The reviewer workflow completed end to end: analysis and evidence projection succeeded, and the same workflow correctly withheld a manuscript whose scientific authority was not established.

EXECUTION

6/6 steps

Completed reviewed plan steps; this is an engineering execution measure.

REGISTERED EVIDENCE

125

Pipeline evidence records projected for this exact run.

REVIEW SURFACES

12 tables / 3 figures

Bounded aggregate browser projections; no patient rows are included.

FORMAL PAPER GATE

WITHHELD AS DESIGNED

The safety boundary prevented technical success from becoming unsupported manuscript authority.

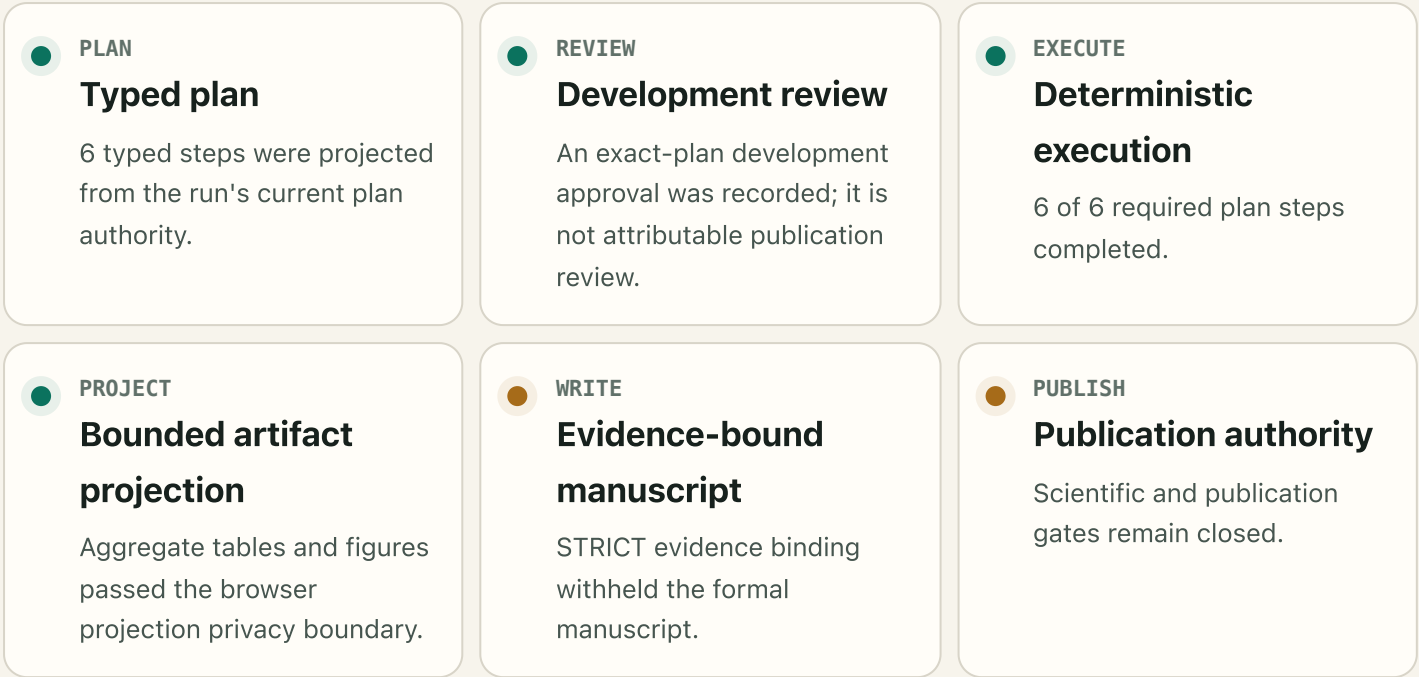
PROVIDER USAGE

14 calls / 162,256 tokens

Recorded estimated cost: \$2.30776.

Authority-aware lifecycle

Green marks a verified workflow boundary. Amber marks an authority intentionally withheld by the safety policy; it is an expected validation outcome, not a failed demonstration.



What the case establishes

Demonstrated

- A typed plan can cross development review and execute all required steps.
- Aggregate tables and figures can be projected without exposing identifier columns or host paths.
- Provider usage, run identity, evidence inventory, and projection inputs remain inspectable.
- A successful analysis does not automatically unlock a manuscript or publication claim.

Outside this demonstration

- Clinical novelty, causal validity, predictive performance, or transportability.
- A finalized publication data seal or externally validated multi-database result.
- Independent attributable scientific review or formal publication activation.
- Superiority over clinicians, conventional workflows, or competing research agents.

Bounded ICU case study

The phenotype analysis is retained as an execution and evidence-binding test case. It is not presented as the system paper's novelty claim.

descriptive_epidemiology

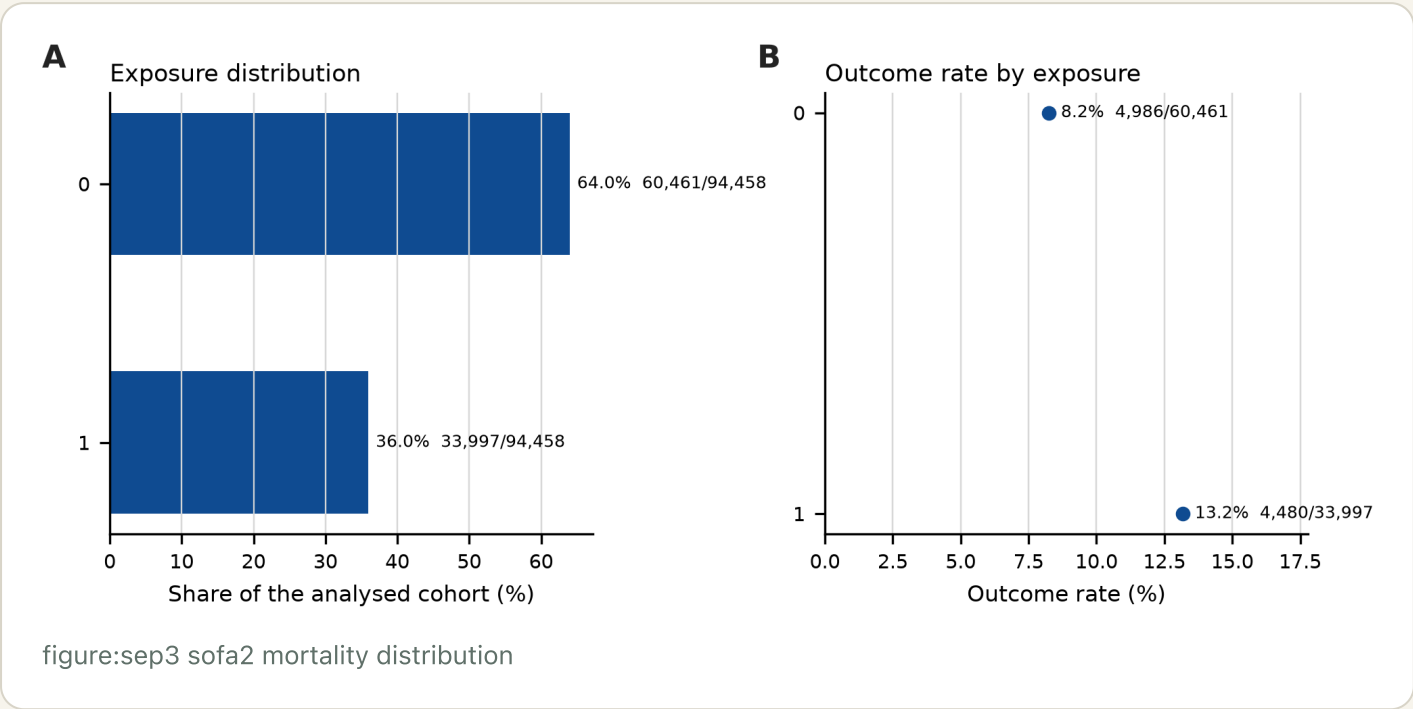
scientific ceiling: descriptive_only

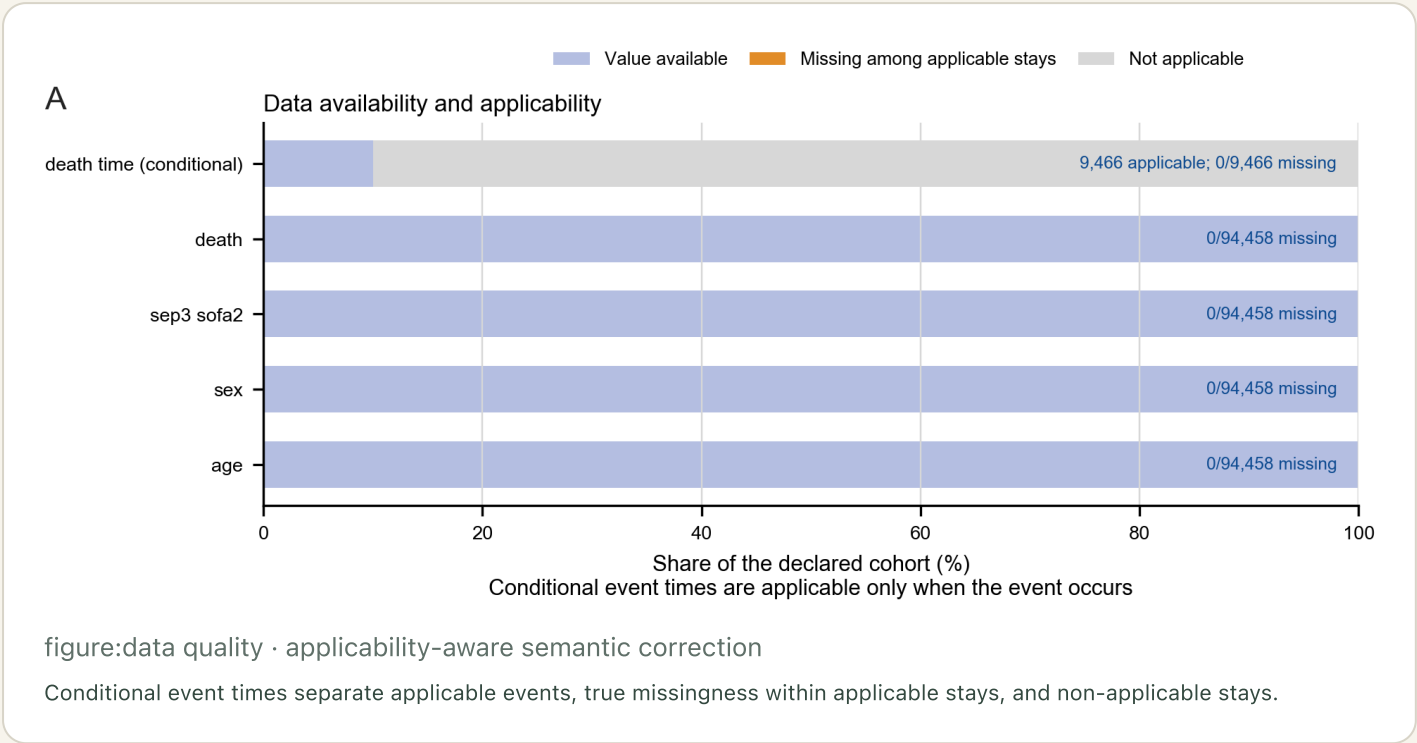
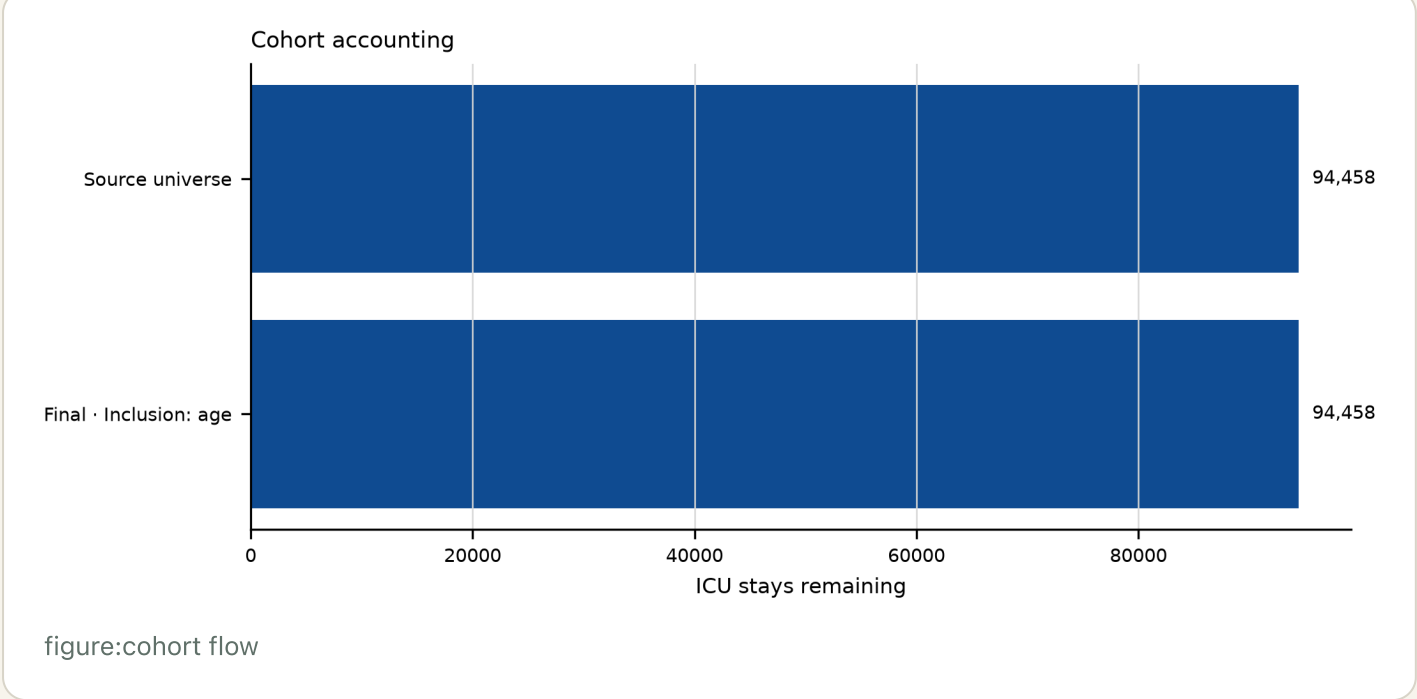
generated_numbers=false

Table exposure_outcome_distribution from step 03_exposure_outcome_distribution.

table_step_artifact_257971d9f784c410

EXPOSURE_LEVEL	N_ROWS	EXPOSURE_DENOMINATOR	EXPOSURE_PCT	OUTCOME_EVENTS	OUTCOME_DENOMINATOR	OUTCOME_RATE_PCT	INTERVAL_METHOD
0.0	60461	94458	64.008342	4986	60461	8.246638	none_counts_only
1.0	33997	94458	35.991658	4480	33997	13.177633	none_counts_only
	94458	94458	100.0	9466	94458	10.021385	none_counts_only





Safety outcome: manuscript authority withheld

The completed reviewer workflow exposes the strongest unresolved scientific requirements instead of laundering them into polished prose.

SEVERITY	FINDING	WHY IT MATTERS	REQUIRED REMEDIATION
BLOCKER	NOVELTY POSITIONING NOT ESTABLISHED analysis	The run has no independent, source-bound comparison showing how its population/setting, exposure/time zero, outcome/estimand, analysis/robustness, data-source transportability, and clinical or methodological contribution differ from retained direct comparators.	Create a comparator matrix from retained source excerpts, record substantive differences on all six dimensions, and obtain independent review. A new database/concept instantiation alone is not a novelty claim.

SEVERITY	FINDING	WHY IT MATTERS	REQUIRED REMEDIATION
BLOCKER	POST BASELINE EXPOSURE TIMING NOT CLOSED analysis	Exposure status is ascertained after time zero, but the final plan does not close exposure opportunity, early events, or a landmark/time-varying alternative.	Pre-specify a non-overlapping follow-up strategy and early death/discharge accounting, or explicitly retain the estimate as descriptive and non-article-grade.
BLOCKER	LITERATURE RETRIEVAL NOT CONDUCTED literature	The bundle contains curated seed references, but no retrieval source completed a search; recency, relevance, and novelty are unverified.	Run a dated database search with inspectable queries and screening receipt, then rebind the exact retained records to the plan.
BLOCKER	INDEPENDENT SCIENTIFIC REVIEW NOT AVAILABLE analysis	No owner-issued independent scientific review receipt is available.	Generate the clinical/methodological reviewer receipt and keep human sign-off separate from the Agent's own review.
BLOCKER	MANUSCRIPT EXACT LITERATURE BINDING INCOMPLETE analysis	The manuscript does not use only exact keys from the run-bound literature bundle.	Regenerate the draft with exact [@key] citations and reject unknown or aggregate-search-only support.
BLOCKER	PAPER AUTHORITY NOT GRANTED manuscript	An evidence-bound draft exists at most; publication authority has not been granted by the Research Agent gates.	Close scientific, display, reporting, provenance, and human-review gates on one exact run authority before external use.

Validation programme needed for a systems paper

1. Add multi-task and multi-database benchmark cases with prespecified success criteria.
2. Compare governed and ungoverned agents on plan drift, hallucination, evidence laundering, and recovery.
3. Run ablations for each authority boundary and report error-detection sensitivity and false blocks.
4. Obtain independent expert review and quantify reproducibility, time, and cost against human workflows.

Projection bindings

Each digest binds the browser-safe projection consumed by this dossier. The report does not read raw patient rows.

SOURCE ARTIFACT AND BINDING SCOPE	SHA-256
agent_plan.json browser projection payload	8adb8bc24cef8ddc5cf08041ef4a15433ac2254a08e74c2ba27e5ff08538a65d
figure_gallery.json browser projection payload	f2276d43b21ea1b92f8c514dc4cafd13981595d3fa96e304b17e32768c8683d9

SOURCE ARTIFACT AND BINDING SCOPE	SHA-256
manuscript_draft.json browser projection payload	31e5aaf92d9ad33aa17cdf3c50a10448bc6d8204b8b05bd0e4d7245f153fe7bb
quality_gate.json browser projection payload	f842386fad936b53f57857c91be5c8b16e3d884102df3074759d3bb0db761c11
result_tables.json browser projection payload	fc63b5a4d8fd137f75123fe4c1ac005da5393c0a476301426baa12987d499776
run_context.json browser projection payload	63bae3a6ae881fd7f9b9929bf7dff9cde3b17bcf35a6829fe0215759a34a2f0f
scientific_readiness.json browser projection payload	a16d5c147b7c9ab7b44b1e4bcdb4fdf656677d6a5bdf2c4650b56e7195d31b4f
source_run_manifest.json browser projection payload	9f613b3c249cb82b2e9607d019220c0c8004681582b6077258f4b544d6ba09e2
system_validation_figure_gallery.json browser projection payload	de987c9d9155e8bee4e1317b669740e3652ddbb17a657888b9732ccda92ca52c
human_review_checkpoint.json run private receipt	0efd235f418ea82153842d3b3fe3e78c32c726848b7dac5a07a11af2cf8f1d0d
provider_hard_stop_ledger.json run private receipt	759ed55918772e42654543e40fc9d951d540fee1ace9c208319ca09a0d663fbd

REVIEWER DEMONSTRATION · ENGINEERING VALIDATION ONLY · NOT A CLINICAL MANUSCRIPT

This deterministic dossier demonstrates the system workflow; it does not grant scientific, clinical, or publication authority to the bounded case study.